

Charles O’Hanlon

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Research Interests

Inductive biases for efficient multimodal learning; world models and learned representations for planning and control; semantic-space prediction (JEPA-family architectures) as an alternative to text-centric scaling; hierarchical structure in learned representations.

Education

New York University, Courant Institute

Expected May 2028

M.S. in Computer Science

Planned coursework: Mengye Ren’s DS-GA 3001 (Advanced Topics in Embodied Learning and Vision); LeCun/Canziani Deep Learning; Bruna mathematical foundations of deep learning.

California Polytechnic State University, San Luis Obispo

Mar 2026

B.S. in Computer Science; Minor in Mathematics

Major GPA: 3.94; GPA: 3.86

Relevant Coursework: Linear Algebra I/II/III, Numerical Analysis, Statistics, Multivariate Statistics, Algorithms, Game Theory, Distributed Computing

Research Experience

Senior Research Assistant — UChicago, Argonne National Lab

Jun 2025 – Sep 2025

advised by Dr. Silvio Rizzi; U.S. Dept. of Energy lab

- Studied Kolmogorov-Arnold Networks as an alternative architectural prior for implicit neural representations of 3D scientific volumes; **+2–15 dB PSNR** over SIREN/MLP baselines.
- Characterized the regimes where KAN’s learned activation functions outperform fixed-activation INRs via systematic ablation on volume frequency content and smoothness; distributed training on the ALCF Sophia DGX A100 cluster.

Undergraduate Research Assistant — Ventura Lab, Cal Poly

Jan 2025 – Jun 2025

advised by Prof. Johnathan Ventura; funded by the National Cancer Institute

- Delivered sit-to-stand repetition counts with **MAE less than 1 repetition** vs. clinician annotations ($n \approx 50$ sessions); compared end-to-end vs. keypoint-based architectures for action segmentation from monocular video.
- Implemented detection/tracking pipelines in **MediaPipe/OpenCV** with models from **OpenMMLab**; built validation harness and error analysis to meet time-boxed protocol requirements.

Undergraduate Research Assistant — AI for Search & Rescue Lab, Cal Poly

Sep 2023 – Dec 2024

advised by Prof. Franz Kurfess; sponsored by Menlo Park Fire Protection District

- Engineered a large-scale **1M+** multimodal dataset from missing person cases for downstream modeling.
- Prototyped multimodal location predictors comparing transformer-based fusion of vision and tabular features against GBDT and agentic baselines; trained/evaluated with **Hugging Face Transformers** and **scikit-learn**.

Posters & Presentations

KAN INRs: Kolmogorov-Arnold Neural Networks for Scientific Volumetric Implicit Neural Representations

Aug 2025

Charles O’Hanlon. Poster. *Learning on the Lawn*, Argonne National Laboratory.

Neural Networks for Missing Persons Location Prediction

Apr 2024

Charles O’Hanlon, Brandon Kim, Ameer Arsala. Poster. *National Missing and Unidentified Persons Conference*.

Industry Experience

ML Engineer Intern — LandingAI

Sep 2025 – Dec 2025

founded by Dr. Andrew Ng; \$57M Series A

- Diagnosed VLM attention anomalies causing prediction truncation via attention-mass visualization; engineered a sequence-seeding intervention resolving **≈90%** of failure cases where anchoring alone failed.
- Conducted parameter-efficient fine-tuning study: 2D sweep over a re-parametrized basis of the 4D LoRA hyperparameter space, exploiting parametrization invariances; **+8%** over prior methods on benchmarks.
- Distilled an anchored **Qwen2-VL** for document content extraction; set new best on internal benchmarks by **≈10%** edit distance.
- Engineered a high-throughput data distillation pipeline labeling **5M+** documents asynchronously with adaptive sampling to balance training datasets within teacher-inference cost constraints.

ML Engineer Intern — FemtoAI

Jun 2023 – Sep 2023

Stanford Brains in Silicon Lab spinout; \$8M Series A

- Improved AI Noise Reduction performance **5%** Δ SI-SDRi by augmenting training with generative samples from FAIR's *AudioGen* targeting low-performing audio categories; an early experiment with generative augmentation as a data-efficiency lever.
- Increased experimentation throughput **10%** by automating train→fine-tune→prune→post-training quantization; orchestrated with **Hydra** and tracked in **Weights & Biases**, enabling production updates to a 16 ms-latency model.
- Deployed an on-prem Linux server and a Python abstraction to evaluate new **PyTorch** models on a custom ML accelerator; reduced pre-production validation friction for the team.

Teaching

Teaching Assistant — CSC480 Artificial Intelligence, Cal Poly

Sep 2025 – Dec 2025

- Led weekly labs and office hours for **50+** students in upper-division Introduction to AI covering heuristic search, constraint satisfaction, knowledge representation, and machine learning.
- Evaluated student implementations of search algorithms, game-playing agents, and ML models, providing technical feedback aligned with course rubric.

Curriculum Design Contributor — CSU AIEIC, Cal Poly SLO

Aug 2025 – Present

- Contributed to the CSU AI Educational Innovations Challenge (\$3M systemwide initiative); co-designed an experiential learning framework where students construct and refine autonomous agents integrated with generative AI tools.

Skills

Languages: Python, C/C++, Cuda

ML: PyTorch, NumPy, Hugging Face Transformers, scikit-learn

Distributed/Infra: CUDA, vLLM, SGLang, Ray, Weights & Biases, Hydra

Vision: OpenCV, MediaPipe, OpenMMLab